14.6

- Tangent plane: $f_x(P_0)(x-x_0)+f_y(P_0)(y-y_0)+f_z(P_0)(z-z_0)=0$
- Normal Line:

Parametric form:

$x=x_0+f_x(P_0)t, \quad y=y_0+f_y(P_0)t, \quad z=z_0+f_z(P_0)t$

Vector form: $[x_0+f_x(P_0)t]\hat{i}+[y_0+f_y(P_0)t]\hat{j}+[z_0+f_z(P_0)t]\hat{k}$

- Plane Tangent to Surface $z=f(x,y)$:

$f_x(P_0)(x-x_0)+f_y(P_0)(y-y_0)-(z-z_0)=0$

Normal $\parallel$: $\hat{n}=[f_x(P_0)(x-x_0)]\hat{i}+[f_y(P_0)(y-y_0)]\hat{j}-(z-z_0)\hat{k}$

- Linearization:

$f(x_0,y_0)\approx L(x,y)=f(x_0,y_0)+f_x(x_0,y_0)(x-x_0)+f_y(x_0,y_0)(y-y_0)$

12.5

- Eq. of Line in Space:

$\underline{x=x_0+v_1t}, \quad \underline{y=y_0+v_2t}, \quad \underline{z=z_0+v_3t}$

$\because \overrightarrow{P_0P}\parallel\vec{v}$

$\because \overrightarrow{P_0P}=\vec{v}t \Rightarrow$

$x\hat{i}+y\hat{j}+z\hat{k}=(x_0\hat{i}+y_0\hat{j}+z_0\hat{k})+(v_1\hat{i}+v_2\hat{j}+v_3\hat{k})t$

• Eq. of Plane in Space:

$Ax + By + Cz = D$, $D = Ax_0 + By_0 + Cz_0$

$\underline{Ax + By + Cz = Ax_0 + By_0 + Cz_0}$

(Normal vector) $\underline{\hat{n} = A\hat{i} + B\hat{j} + C\hat{k}}$ or $\hat{n} = (A, B, C)$

$$r(t) = r_0 + t\,|\vec{v}|\,\hat{v} \qquad \left(\hat{v} = \frac{\vec{v}}{|\vec{v}|}\right)$$

Final position (pt) ↑ $r(t)$; Initial position ↑ r_0; Time ↑ t; Speed ↑ $|\vec{v}|$; Direction ↑ $\hat{v}$

Distance = Time x speed

Distane from Point (S) to Line throug P:

$$\boxed{d = \frac{|\overline{PS} \times \vec{v}|}{|\vec{v}|}}$$

Gradient Descent Method

• $X_{n+1} = X_n - \lambda_n S_n$

• $S_n = \nabla f(X_n) = \begin{bmatrix} f_x(x_n, y_n) \\ f_y(x_n, y_n) \end{bmatrix}$

• $H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$ → If variables occur, put $X_n(x_n, y_n)$

• $\lambda_n = \dfrac{S_n^t S_n}{S_n^t H S_n}$